JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Affiliated to DBATU, RTMNU & MSBTE, Mumbai

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Education to Eternity

An Autonomous Institute, with NAAC "A" Grade

Department of Computer Science & Engineering

"A place to learn; A chance to grow"

Session: 2024-25

MID SEMESTER EXAMINATION 2

Course: B. Tech. in Computer Science & Engineering **Year/Semester:** 2nd year/ 4th Sem

Subject : Design & Analysis of Algorithm **Subject Code:** CS4T002

Max Marks: 20 **Date:** 15/04/2025 **Duration:-** 1 Hr.

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

Students should be able to:

CO1: Recall basic concepts of algorithm in analysis and Design of algorithms.
CO2: Examine Recurrence relations, solutions of recurrence of searching sorting methods.
CO3: Apply Greedy methods used for analysis and Design of Algorithm.
CO4: Analyse Dynamic Programming concepts in designing algorithm.
CO5: Evaluate advanced techniques and tools available for algorithm analysis and development.

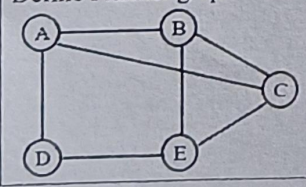
SECTION A

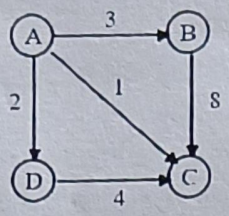
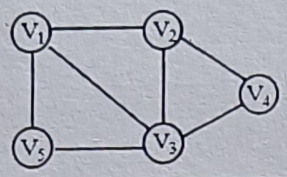
1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Longest Common Subsequence with example.	L1/CO4	1
Q.2	Write the principle of All Pairs shortest Path Algorithm.	L1/CO4	1
Q.3	Describe Graph Colouring with example.	L1/CO4	1
Q.4	List out the examples of Dynamic Programming Stategy.	L1/CO4	1

SECTION B

1. **All Questions are compulsory.**
2. **Answer in two or three sentence.**

Q.5	State the principle of Backward and Forward Multistage Graph.	L1/CO4	2
Q.6	Differentiate between Dynamic Programming and Divide & Conquer Strategy	L4/CO4	2
Q.7	Give two solutions for 4-Queen Problem and 8-Queen Problem	L2/CO4	2
Q.8	Define Planner graph? Convert the given graph into planner graph. 	L1/CO5	2

SECTION C			
1. Solve any two questions.			
Q.9	Implement Floyd – Warshall algorithm on for the following graph. 	L3/CO4	4
Q.10	Determine LCS of X = EXPONENTIAL Y = POLINOMIAL Write recurrence relation for LCS.	L3/CO4	4
Q.11	Apply Graph Colouring on the given graph. State the total number of possible solutions by using solution space tree. 	L3/CO5	4